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Experiment: Jenkins Installation and Pipeline Execution

LOs mapped: LO1 and LO3

1. Aim

To install and configure Jenkins and demonstrate the execution of Declarative and Scripted Jenkins Pipelines.

2. Objectives

- To install Jenkins using the Jenkins WAR file.
- To configure Jenkins on a local system.
- To create a Jenkins Pipeline project.
- To execute a Declarative Pipeline.
- To execute a Scripted Pipeline.
- To verify successful pipeline execution through the Jenkins Console Output.

3. Software/Hardware Requirements

Hardware:

- Computer/Laptop
- Minimum 4 GB RAM recommended
- Internet connection

Software:

- Windows Operating System
- Java Development Kit (JDK)
- Jenkins
- Web browser such as Google Chrome or Microsoft Edge

4. Theory

Jenkins is an open-source automation server used to automate various stages of software development, including building, testing, and deployment.

Jenkins supports Continuous Integration (CI) and Continuous Delivery/Deployment (CD). It can automatically execute predefined tasks whenever changes are made to a software project.

A Jenkins Pipeline defines a sequence of automated processes that Jenkins executes.

Pipelines are generally written using Groovy-based syntax.

There are two commonly used Pipeline syntaxes:

Declarative Pipeline

A Declarative Pipeline provides a structured and simplified syntax for defining a Jenkins Pipeline.

Basic structure:

```
pipeline {
   agent any

   stages {
      stage('Hello') {
         steps {
            echo 'Hello, Jenkins!'
         }
```

```
        }
    }
}
```

Important components:

- pipeline – Defines the Pipeline.
- agent – Specifies where the Pipeline should execute.
- stages – Contains the different stages of the Pipeline.
- stage – Represents an individual stage.
- steps – Contains the commands executed within a stage.
- echo – Displays a message in the Jenkins console.

Scripted Pipeline

A Scripted Pipeline provides greater flexibility and uses Groovy scripting directly.

Example:

```
node {
    stage('Hello') {
        echo 'Hello, Jenkins!'
    }
}
```

Here, node allocates a Jenkins executor/workspace and the stage defines a particular stage of execution.

5. Procedure

Part A – Jenkins Installation

1. Open Command Prompt.
2. Verify that Java is installed:

```
java -version
```

3. Navigate to the directory containing the Jenkins WAR file:

```
cd "%USERPROFILE%\Downloads"
```

4. Verify the Jenkins WAR file:

```
dir jenkins.war
```

5. Start Jenkins using:

```
java -jar jenkins.war
```

6. Wait until the following message appears:

```
Jenkins is fully up and running
```

7. Open a web browser and navigate to:

```
http://localhost:8080
```

8. Enter the initial administrator password displayed in the Command Prompt.
9. Select Install suggested plugins.
10. Create the Jenkins administrator account.
11. Complete the setup and open the Jenkins Dashboard.

6. Procedure for Declarative Pipeline

1. Open the Jenkins Dashboard.
2. Click New Item.
3. Enter the project name:

Hello-Jenkins

4. Select Pipeline.
5. Click OK.
6. Scroll to the Pipeline section.
7. Select Pipeline script as the definition.
8. Enter the following Pipeline:

```
pipeline {
   agent any

   stages {
      stage('Hello') {
         steps {
            echo 'Hello, Jenkins!'
         }
      }
   }
}
```

9. Click Save.
10. Click Build Now.
11. Open the generated build number.
12. Select Console Output.
13. Verify that the message is displayed and the build finishes successfully.

Expected output:
Hello, Jenkins!
Finished: SUCCESS

7. Procedure for Scripted Pipeline

1. Return to the Jenkins Dashboard.
2. Click New Item.
3. Enter the project name:

Scripted-Jenkins

4. Select Pipeline.
5. Click OK.
6. Scroll to the Pipeline section.
7. Enter the following Scripted Pipeline:

```
node {
   stage('Hello') {
      echo 'Hello, Jenkins!'
   }
}
```

8. Click Save.
9. Click Build Now.
10. Open the generated build number.
11. Select Console Output.
12. Verify the execution result.

Expected output:
Hello, Jenkins!
Finished: SUCCESS

8. Result
Jenkins was successfully installed and configured on the local system. Both Declarative and Scripted Jenkins Pipelines were created and executed successfully. The output was verified using the Jenkins Console Output, and the builds completed with the status:
Finished: SUCCESS

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```
C:\Users\15L>java -version
java version "25.0.2" 2026-01-20 LTS
Java(TM) SE Runtime Environment (build 25.0.2+10-LTS-69)
Java HotSpot(TM) 64-Bit Server VM (build 25.0.2+10-LTS-69, mixed mode, sharing)

C:\Users\15L>dir "%USERPROFILE%\Downloads\jenkins.war"
 Volume in drive C is Windows
 Volume Serial Number is 2246-7C0D

 Directory of C:\Users\15L\Downloads

01/27/2026  03:23 PM        96,260,165 jenkins.war
               1 File(s)     96,260,165 bytes
               0 Dir(s)  761,837,002,752 bytes free

C:\Users\15L>cd "%USERPROFILE%\Downloads"

C:\Users\15L\Downloads>dir jenkins.war
 Volume in drive C is Windows
 Volume Serial Number is 2246-7C0D

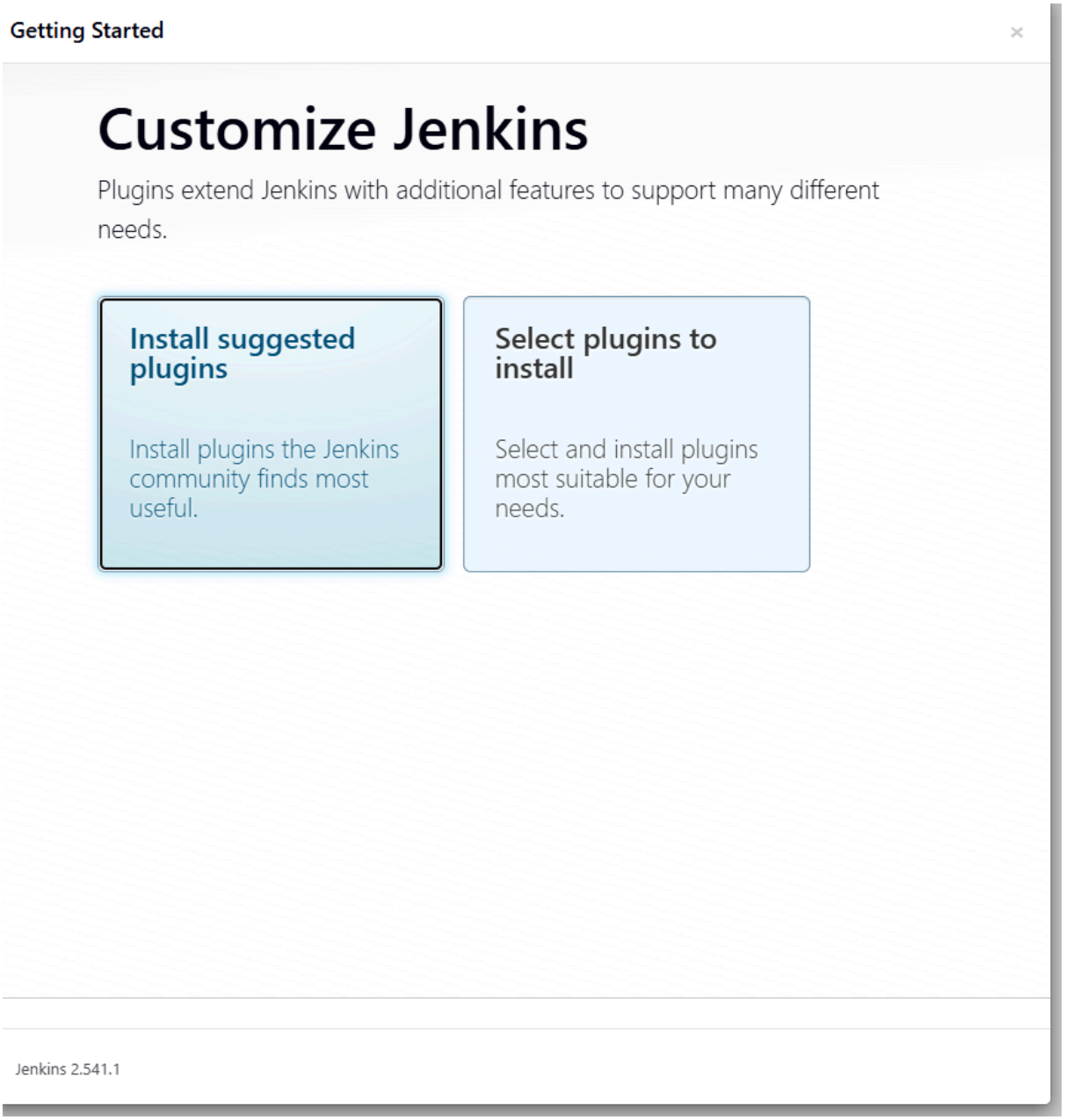
 Directory of C:\Users\15L\Downloads

01/27/2026  03:23 PM        96,260,165 jenkins.war
               1 File(s)     96,260,165 bytes
               0 Dir(s)  761,838,243,840 bytes free
```

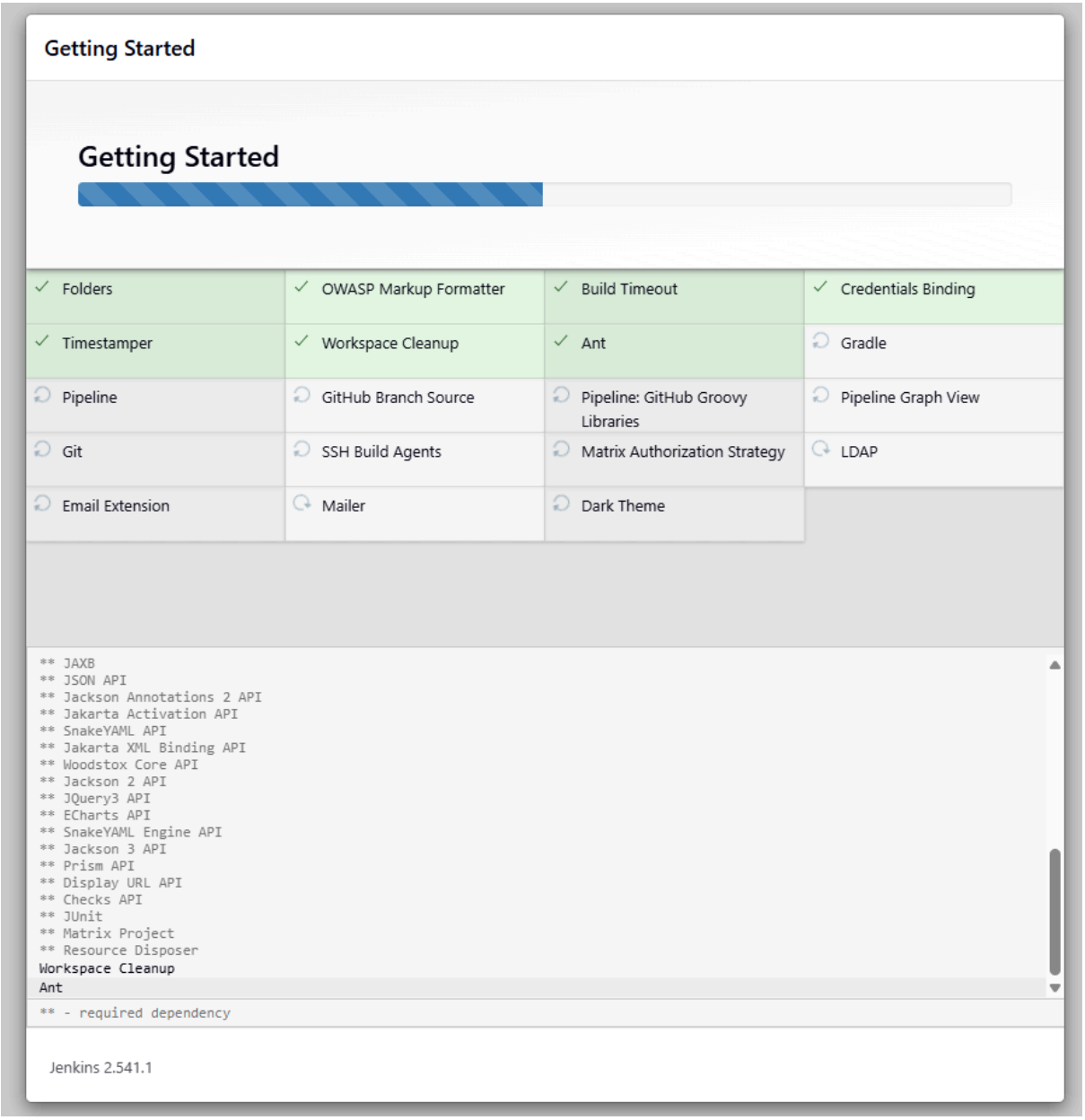
```
C:\Users\15L\Downloads>java -jar jenkins.war
Running from: C:\Users\15L\Downloads\jenkins.war
webroot: C:\Users\15L\.jenkins\war
2026-08-07 07:53:55.017+0000 [id=3]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-08-07 07:53:57.754+0000 [id=3]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-08-07 07:53:57.855+0000 [id=3]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.5; built: 2025-12-03T22:18:24.732Z; git: 4905d09d7b6801e792d7b73946cf7b66e4be25d6; jvm 25.0.2+10-LTS-69
2026-08-07 07:53:58.969+0000 [id=3]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-08-07 07:53:59.076+0000 [id=3]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::objectFieldOffset has been called by com.thoughtworks.xstream.converters.reflection.SunUnsafeReflectionProvider (file:/C:/Users/15L/.jenkins/war/WEB-INF/lib/xstream-1.4.21.jar)
WARNING: Please consider reporting this to the maintainers of class com.thoughtworks.xstream.converters.reflection.SunUnsafeReflectionProvider
WARNING: sun.misc.Unsafe::objectFieldOffset will be removed in a future release
2026-08-07 07:53:59.760+0000 [id=3]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: C:\Users\15L\.jenkins found at: $user.home/.jenkins
2026-08-07 07:53:59.955+0000 [id=3]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@61c76850{Jenkins v2.541.1,/,b=file:///C:/Users/15L/.jenkins/war/,a=AVAIL
ABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@702424f38{STARTED}}
2026-08-07 07:54:00.021+0000 [id=3]     INFO    o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@3721177d{HTTP/1.1, (http/1.1)}{0.0.0.0:8080}
2026-08-07 07:54:00.023+0000 [id=3]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@14e2e1c3{STARTING}[12.1.5,sto=0] @2171ms
2026-08-07 07:54:00.025+0000 [id=67]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-08-07 07:54:00.181+0000 [id=57]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.541.1
2026-08-07 07:54:00.314+0000 [id=75]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-08-07 07:54:00.404+0000 [id=95]    INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-08-07 07:54:01.606+0000 [id=104]   INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-08-07 07:54:01.614+0000 [id=108]   INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-08-07 07:54:01.615+0000 [id=88]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-08-07 07:54:01.979+0000 [id=80]    INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-08-07 07:54:01.981+0000 [id=92]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-08-07 07:54:01.982+0000 [id=98]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-08-07 07:54:01.984+0000 [id=97]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-08-07 07:54:02.056+0000 [id=83]    INFO    jenkins.install.SetupWizard#init:
[CRLF]>
[CRLF]> ************************************************************
[CRLF]> ************************************************************
[CRLF]> ************************************************************
[CRLF]>
[CRLF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[CRLF]> Please use the following password to proceed to installation:
[CRLF]>
[CRLF]> dc66cbb9470a469cb2413572893734b9
[CRLF]>
[CRLF]> This may also be found at: C:\Users\15L\.jenkins\secrets\initialAdminPassword
[CRLF]>
[CRLF]> ************************************************************
[CRLF]> ************************************************************
[CRLF]> ************************************************************

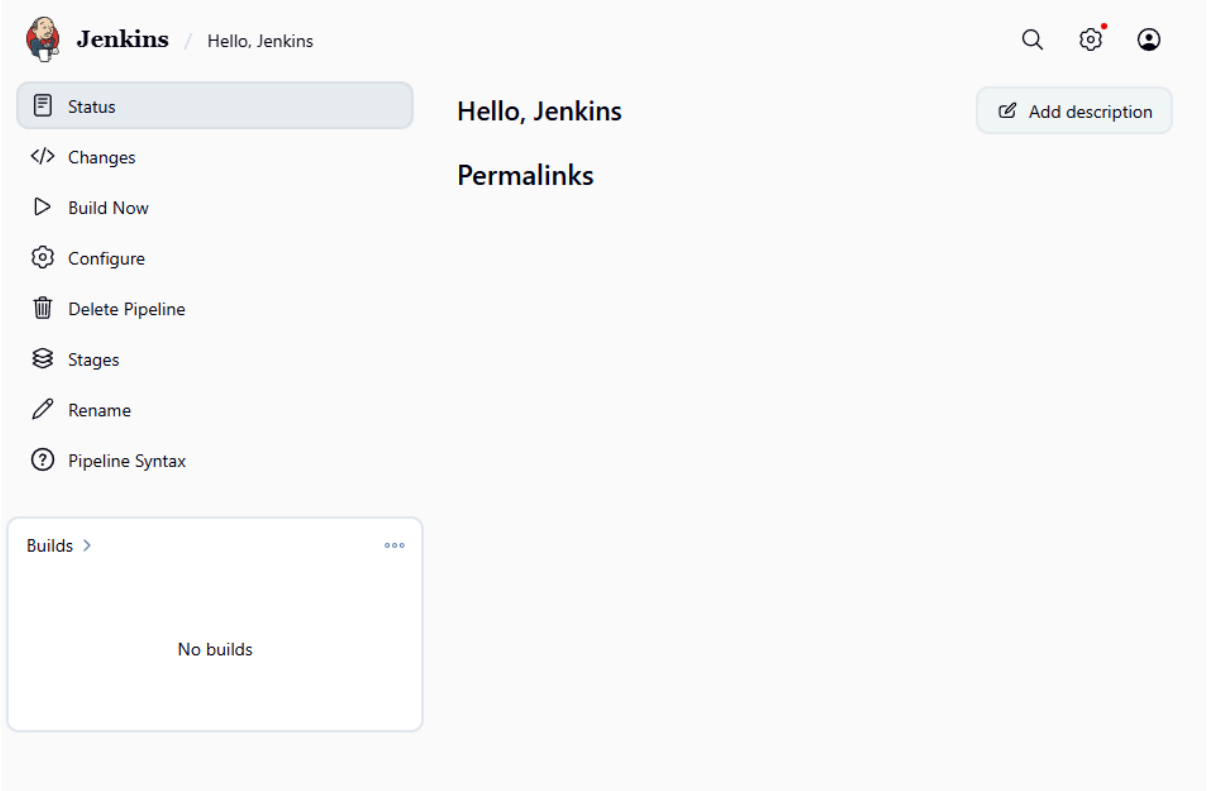
2026-08-07 07:54:06.447+0000 [id=83]    INFO    jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-08-07 07:54:06.481+0000 [id=57]    INFO    hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by com.sun.jna.Native in an unnamed module (file:/C:/Users/15L/.jenkins/war/WEB-INF/lib/jna-5.18.1.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled
```

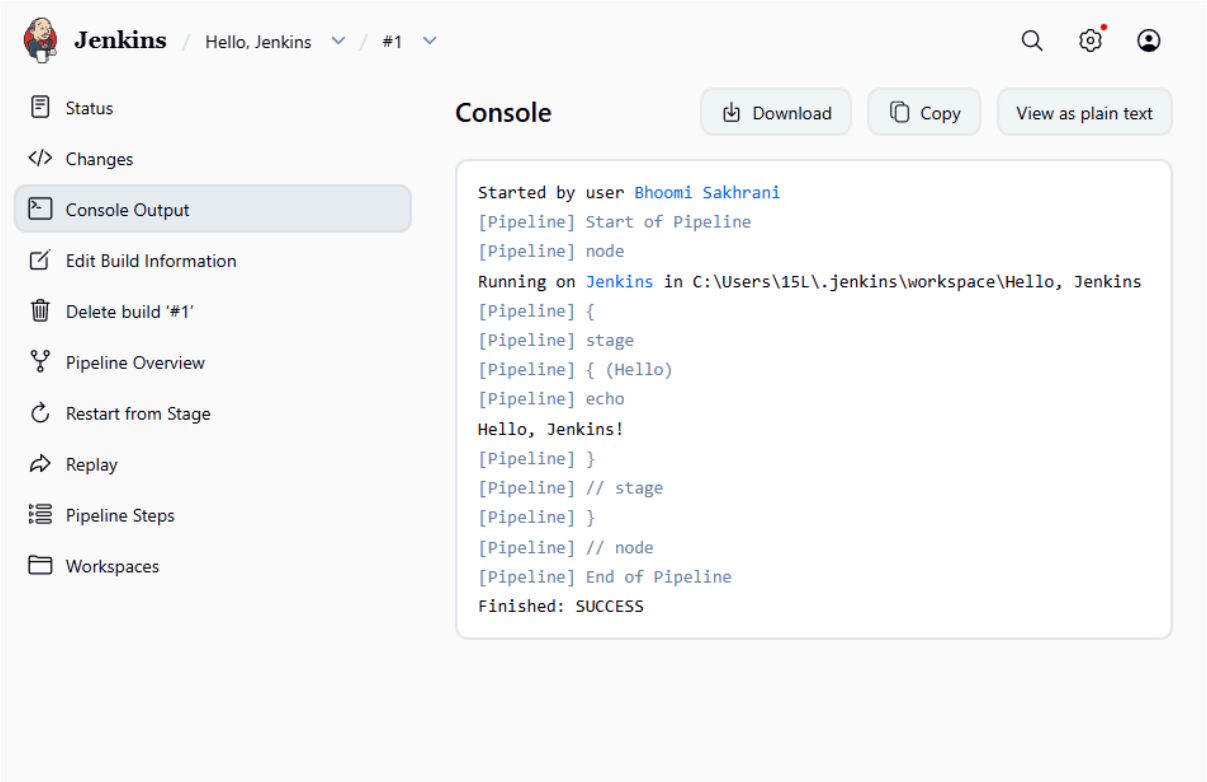
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9. Conclusion

Thus, Jenkins was successfully installed and configured, and the working of Declarative and Scripted Pipelines was demonstrated. The experiment provided an understanding of Jenkins automation and the basic structure and execution of Jenkins Pipelines.